

Gaurav Ghop

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AI Software Engineer with over 3 years of experience building production LLM systems, multimodal ML pipelines, and scalable backend services.

U.S. work authorized (EAD) — no sponsorship required.

TECHNICAL SKILLS

Languages: Python, C++, Java, SQL

AI & Machine Learning: PyTorch, Hugging Face Transformers, Computer Vision, Deep Learning, CNNs, Transformers, Classification, Segmentation

LLMs & Retrieval: RAG, LangChain, LangGraph, Vector Search (FAISS, pgvector), LoRA Fine-tuning, LLM Evaluation, AWS Bedrock, MCP

GPU & Inference: CUDA, Triton, TensorRT, NVIDIA Jetson, Quantization (INT8/FP8), KV Cache, Edge AI

Backend & Infrastructure: FastAPI, Kafka, PostgreSQL, Docker, Kubernetes, AWS, Linux

EXPERIENCE

Tiny Archives

Jun 2025 – Aug 2025

AI Software Engineer, Summer Intern

New York, NY

- Cut LLM API expense by **28%** with a confidence-based router that handled routine extractions on a locally hosted **Qwen 2.5 7B** and escalated ambiguous cases to **GPT-4o**, backed by **prompt caching** and automated fallback.
- Shipped a **RAG** pipeline over mixed archival formats (EAD/XML, DOCX, CSV, PDF) with **LangChain** and **pgvector**, combining hybrid retrieval and reranking, grounding responses with document-level citations.

NYU Steinhardt, Play & Language Lab

Dec 2024 – May 2025

Machine Learning Research Assistant

New York, NY

- Automated annotation of infant vocalizations and caregiver speech across **500 hours** of home audio with **PyTorch** classification and segmentation models, cutting manual coding time by **40%** at **90%** agreement with human coders.
- Engineered ingestion and validation tooling for the lab's audio/video corpus (**Databrary**, later licensed to **Meta**), flagging misaligned timestamps and inconsistent labels, reducing hours of manual review.

Duende Ingenious

Jul 2023 – Aug 2024

Founding Engineer

Pune, India

- Designed an event-driven **feature ingestion pipeline** with **Kafka**, **SQS**, and asynchronous **Python (asyncio)** workers, producing reliable training and inference datasets while using retries (with back-off) and dead-letter recovery.
- Developed a real-time **dispatch-ranking model** in **Python** that ranked delivery partners using rider location, pricing, historical completion rate, and traffic features, improving **on-time deliveries by 18%** over rule-based assignment.

EQ Technologic

Aug 2021 – Jun 2023

Software Engineer, Data & Platform

Pune, India

- Developed **Python** data pipelines on an enterprise data-virtualization platform, federating PLM, ERP, and MES sources into a **common data model** exposed over **REST/ODATA** API's across defense deployments (**incl. Lockheed Martin, US Navy**).
- Maintained sync reliability across **Kafka** consumers with idempotent processing and automated reconciliation, adding data-profiling checks that caught schema drift at ingest and cut defect triage time by **18%**.

PROJECTS

Formulary Analysis, Local LLM Fine-Tuning Pipeline | *MLX-LM, LoRA, Llama 3.2 3B, PostgreSQL* | [LINK](#)

- Fine-tuned a Llama 3.2 3B with **LoRA** over an offline pipeline ingesting formulary PDFs, reaching **82%** domain accuracy at **41 tok/s** under **4 GB** RAM.

Sentari, Edge-AI Traffic-Violation Detection (co-founded) | *Jetson Nano, YOLO, ByteTrack, TensorRT* | [LINK](#)

- Shipped a real-time on-device detector (fine-tuned **YOLO**, **ByteTrack**, **ALPR**) for helmetless, triple-seating, and wrong-way riding, taking it from **4 to 15 FPS** with INT8 quantization, pruning, and **TensorRT**, across **20M frames** in production.

Shadow Engineer, Onboarding Agent | *LangGraph, MCP, AWS Bedrock, RAGAS* | [LINK](#)

- Designed a **LangGraph** agent that plans against a ticket and pulls context from Jira, the repo, and commit history through **MCP** servers, running a plan-act-observe loop with checkpointed state and retries on failed **tool calls**.

EDUCATION

New York University

Sept 2024 – May 2026

M.S. Computer Engineering

New York, NY

Relevant coursework: High-Performance Machine Learning, Distributed Systems, Machine Learning Operations (MLOps).

Vishwakarma Institute of Information Technology

Aug 2017 – May 2021

B.Tech. Electronics & Telecommunications Engineering

Pune, India

ACHIEVEMENTS

Merit Scholarship for graduate tuition, NYU • **Runner-Up** AWS GenAI Hackathon 2025 • **Finalist** Google DeepMind Hackathon 2025 • **Runner-Up** Smart India Hackathon 2020 • ACM ICPC Regionals 2019 • CodeChef 4★